Article

# Toward Interpretable Voice-Based Parkinson's Disease Screening via Joint Transfer Function-Feature-Classifier Optimization

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## 1. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder worldwide, arising from progressive degeneration of dopaminergic neurons in the substantia nigra [1]. Its global burden has risen sharply alongside population aging: an estimated 3.15 million individuals were living with PD in 1990, a number that grew to 11.8 million by 2021 and is projected to reach 25.2 million by 2050 [2,3]. Diagnosis remains primarily clinical, relying on the observation of cardinal motor signs, namely bradykinesia, rigidity, resting tremor, and postural instability, under established frameworks such as the Movement Disorder Society (MDS) Clinical Diagnostic Criteria [4], typically supported by assessment of the patient's response to levodopa therapy. This process is inherently subjective and is particularly prone to misdiagnosis at early disease stages, when motor signs may be mild, intermittent, or absent [5]. Given that pharmacological and neuroprotective interventions achieve greatest efficacy when initiated before extensive neuronal loss [6], there exists a pressing clinical need for objective, low-cost, and non-invasive markers capable of enabling earlier and more consistent detection than motor examination alone.

Among candidate biomarkers, vocal impairment has attracted particular attention. Hypokinetic dysarthria is estimated to affect the large majority of individuals with PD [7,8], arising from rigidity and reduced range of motion in the laryngeal, respiratory, and articulatory musculature. These physiological changes produce measurable perturbations in the acoustic signal, namely increased cycle-to-cycle variation in fundamental frequency (jitter), increased cycle-to-cycle variation in amplitude (shimmer), a reduced harmonics-to-noise ratio (HNR) reflecting incomplete glottal closure, and altered nonlinear measures of vocal-fold vibration such as recurrence period density entropy and detrended fluctuation analysis [9]. Such vocal changes can emerge early in the disease course, at times preceding or co-occurring with the earliest overt motor signs by as much as a decade [8], and, unlike gait or tremor assessment, can be captured remotely, repeatedly, and inexpensively using nothing more than a standard microphone. This renders voice analysis a particularly attractive candidate for scalable, low-burden screening, including telemedicine and home-monitoring scenarios that cannot readily accommodate in-person motor examination.

Motivated by this, a substantial and rapidly growing body of work has applied machine learning to acoustic features extracted from sustained-phonation or connected-speech recordings for automated PD detection [10]. Given that the number of extractable acoustic measures is often large relative to the number of available recordings, feature selection is commonly employed to reduce dimensionality, limit overfitting, and improve interpretability. Wrapper approaches built around population-based metaheuristic algorithms have become an increasingly popular choice for this task, owing to their derivative-free search mechanisms and their capacity to explore high-dimensional, non-convex feature spaces without requiring restrictive assumptions on the underlying data distribution.

Despite this progress, several limitations persist across the existing literature. First, the large majority of studies treat the choice of classifier as fixed and predetermined, optimizing only the feature subset; the complementary question of which classifier, or combination of classifiers, best exploits a given feature subset is rarely optimized jointly within the same search process. Even recent work that does optimize feature selection and a classifier together typically keeps the classifier type itself fixed and tunes only its hyperparameters jointly with the feature subset [11], rather than searching over which classifier to use in the first place. This separation implicitly assumes that an optimal feature subset is invariant to the classifier that will ultimately consume it, an assumption that is not generally supported, since different classifiers exploit different aspects of the feature space and may therefore favor markedly different feature subsets. Second, feature selection and classifier selection are commonly performed sequentially rather than jointly, such that suboptimal decisions made at the feature-selection stage cannot be corrected during classifier selection, and vice versa; this two-stage pipeline risks converging to a solution that is locally optimal for each subproblem but jointly suboptimal overall, a pattern still evident in very recent PD-specific pipelines that pair a filter-based feature-selection stage with a separately optimized classification ensemble [12]. Third, and less widely recognized, the transfer function that converts a metaheuristic's continuous search space into the binary feature masks it actually evaluates is itself treated as a fixed, externally chosen design decision rather than as something the search can determine: the S-shaped and V-shaped families remain the default convention across the literature, yet comparative evaluations spanning multiple transfer functions on the same task consistently find that no single choice dominates across datasets and algorithms [13], such that this too is a design choice usually settled by ad-hoc pre-selection or trial-and-error rather than by the optimization itself, alongside the feature subset and classifier it jointly searches over. Taken together, these limitations motivate a unified search framework capable of jointly optimizing feature subsets, classifier configurations, and the transfer function governing binarization, one that is validated across multiple, independent datasets to ensure robustness and generalizability of the resulting diagnostic model.

Therefore, this study proposes a joint transfer function-, feature-, and classifier-optimization pipeline for voice-based PD screening, built on the plain Competitive Swarm Optimizer (CSO). The main contributions of this research are as follows:

- We propose a novel paradigm for joint search over the acoustic feature subset, the classifier that exploits it, and the transfer function used to binarize the search itself, within a single optimization encoding, rather than optimizing feature selection alone or fixing a binarization scheme in advance.
- We extend this joint encoding with a per-particle transfer-function-selector segment, so a plain Competitive Swarm Optimizer searches over which of five candidate binarization schemes governs each particle, rather than every particle in the population sharing one transfer function fixed before the search begins.
- We compare the proposed method against seven recently published metaheuristic feature-selection approaches applied specifically to PD datasets, under one shared encoding, fitness function, and evaluation protocol, thereby demonstrating its competitiveness.
- We examine the acoustic features most consistently selected by the proposed pipeline in light of established physiological correlates of parkinsonian dysphonia, so as to assess the biological plausibility and potential clinical interpretability of the resulting screening model.

The remainder of this paper is organized as follows. Section 2 reviews related work on PD diagnosis, voice biomarkers, and metaheuristic feature selection. Section 3 describes the datasets employed and the proposed methodology. Section 4 presents the experimental results, including comparative performance evaluation and statistical significance testing. Section 5 discusses the clinical implications and limitations of the proposed approach. Section 6 concludes the paper.

## 2. Related Work

### 2.1. *Current Diagnostic Practice and the Case for Objective Biomarkers*

Clinical diagnosis of PD remains anchored to the MDS Clinical Diagnostic Criteria [4]: a judgment, made by a trained examiner, of whether bradykinesia is present together with rest tremor and/or rigidity, refined by supportive features and exclusion criteria and typically corroborated by the patient's response to levodopa therapy. Because this judgment rests on the visual and manual assessment of motor signs that can be subtle, intermittent, or altogether absent early in the disease course, diagnostic accuracy is known to vary considerably with examiner experience and disease duration, and misdiagnosis remains common enough at early stages to be a recurring concern in the movement-disorders literature [5]. This has motivated a broader search for objective biomarkers that do not depend on a clinician's subjective interpretation of motor signs. Dopaminergic imaging (DAT-SPECT, PET), cerebrospinal-fluid and blood-based $\alpha$-synuclein assays, genetic testing, and, more recently, wearable and other digital-sensing modalities have all been investigated as candidate objective markers [14]; each can support diagnosis or staging, but each also carries some combination of high cost, invasiveness, specialized equipment, or limited accessibility outside tertiary care centers [14]. Voice and speech recordings, by contrast, can be acquired with nothing more than a standard microphone, at negligible marginal cost, and without any physical contact with the patient, a combination of properties that renders them an unusually attractive candidate among the emerging digital-biomarker modalities, and the specific focus of this study.

### 2.2. *Acoustic and Speech Biomarkers of Parkinsonian Dysphonia*

Hypokinetic dysarthria, characterized by reduced vocal loudness, monopitch, imprecise articulation, and breathy or hoarse voice quality, is estimated to affect the large majority of individuals with PD at some point in the disease course [7,8]. Its physiological basis lies in rigidity and bradykinesia affecting the laryngeal, respiratory, and articulatory musculature: reduced vocal-fold adduction force and control lead to incomplete or irregular glottal closure, reduced respiratory support lowers subglottal pressure and vocal intensity, and reduced range of motion in the articulators blurs consonant and vowel targets [9]. These physiological changes are reflected in several well-established acoustic correlates, namely increased cycle-to-cycle perturbation in fundamental frequency (jitter) and amplitude (shimmer), reduced harmonics-to-noise ratio (HNR) reflecting incomplete glottal closure, reduced vowel space area reflecting imprecise articulation, and altered nonlinear dynamical measures of vocal-fold vibration, including recurrence period density entropy (RPDE), detrended fluctuation analysis (DFA), and pitch period entropy (PPE), that are more sensitive than classical perturbation measures to the aperiodic, weakly chaotic component of parkinsonian phonation [9]. Given that these physiological changes can begin before overt motor signs are clinically apparent, acoustic and broader speech biomarkers have increasingly been proposed as tools for prodromal detection and longitudinal monitoring, not merely post-diagnosis screening [8,15]. This physiological grounding is revisited in Section 5 when interpreting which acoustic features the proposed pipeline selects most consistently.

### 2.3. *Metaheuristic Feature Selection for PD Detection: Progress and Methodological Gaps*

Wrapper-based feature selection couples a search procedure with a downstream classifier's cross-validated performance as the objective to optimize. Population-based metaheuristics, being derivative-free and requiring no assumption of a smooth or convex fitness landscape, have become a standard choice of search procedure for this task across biomedical applications generally [16]. Within voice-based PD detection specifically, a fast-growing line of work applies swarm- and evolution-inspired algorithms to select acoustic feature subsets ahead of a downstream classifier, most commonly encoding the search space as a binary mask over the candidate feature set: particle-swarm-, butterfly-, and whale-inspired variants [17], particle swarm optimization jointly tuning the feature subset and classifier hyperparameters [18], a modified grey wolf optimizer [19], a modified hunger games search [20], a quantum mayfly optimizer [21], a genetic algorithm with filter-based pre-selection [22], a beluga whale optimizer [23], and hybrids chaining two algorithms in sequence, namely a whale-to-grey-wolf cascade [24] and a chaotic grey wolf-dragonfly cascade [25].

As discussed in Section 1, the classifier consuming the selected features is fixed and predetermined in the large majority of this work, rarely optimized jointly with the feature subset within a single search process. Even where classifier hyperparameters are tuned alongside the feature subset [18], the classifier type itself is still chosen in advance rather than searched over. This omission is not merely a matter of scope: because different classifiers weight, combine, and interact with input features in fundamentally different ways, the feature subset best suited to one classifier need not be the one best suited to another, so fixing the classifier a priori risks discarding feature subsets that would perform poorly under the assumed classifier but well under an alternative one that was never considered. Extending the search to include the classifier itself, however, is not a trivial enlargement of the existing formulations; it substantially increases the dimensionality and, more importantly, the combinatorial complexity of the search space, since feature and classifier choices interact rather than contribute independently to fitness, which is precisely the setting in which premature convergence and loss of population diversity become most damaging to search quality. This is the gap that the encoding proposed in this study is designed to close, by searching a higher-dimensional, joint feature-and-classifier space without sacrificing search quality relative to feature-selection-only formulations.

## 3. Materials and Methods

### *3.1. Datasets and Study Population*

Two public, retrospective, previously collected acoustic datasets are used, which are presented as follows.

**Oxford Parkinson's Disease Voice Dataset**[1] [26] comprises 195 sustained phonation (vowel /a/) recordings from 32 subjects, of whom 147 recordings (75.4%) are labeled Parkinsonian and 48 (24.6%) healthy control, that is, class-imbalanced at the recording level. Each recording is described by 22 acoustic features, including fundamental-frequency perturbation measures (jitter variants), amplitude perturbation measures (shimmer variants), harmonics-to-noise ratio, and nonlinear dynamical measures (RPDE, DFA, spread1/2, D2, PPE). Subjects contribute multiple recordings each, approximately six on average, which is the reason subject-grouped cross-validation is required rather than a plain record-level split.

**Naranjo Replicated Acoustic Features Dataset**[2] [27] comprises 240 recordings from 80 subjects (40 Parkinsonian, 40 healthy controls, perfectly class-balanced at the subject level), each subject contributing exactly 3 repeated recordings. Each recording is described by 45 acoustic features (jitter, shimmer, and amplitude-perturbation variants together with a gender covariate), extracted specifically to support test-retest reliability analysis across repeated recordings of the same speaker. This renders subject-grouped evaluation especially important for this dataset, since its very design guarantees near-identical repeated measurements per subject.

### *3.2. Data Preprocessing and Subject-Grouped Cross-Validation*

No missing values are present in either dataset; features are used in their raw scale, with per-fold standardization applied inside each classifier's own pipeline rather than globally, so as to avoid leaking validation-fold statistics into training. Every fitness evaluation performed by every optimizer in this study uses 5-fold cross-validation, grouped by subject identifier and stratified by class label. This guarantees that all recordings from

[1] https://archive.ics.uci.edu/dataset/174/parkinsons
[2] https://archive.ics.uci.edu/dataset/489/parkinson+dataset+with+replicated+acoustic+features

a given subject fall entirely within a single fold, such that no classifier ever trains on one recording from a subject while being validated on another recording from that same subject. Out-of-fold predictions from all 5 folds are concatenated before any performance metric is computed, such that every reported score is computed over the entire dataset under a strict train/validation separation, rather than averaged across fold-level scores.

### *3.3. Transfer Function-, Feature-, and Classifier- Joint Optimization Encoding*

Every optimizer in this study searches over a shared, particle-based encoding: a continuous position vector of length $5 + D + 5$, where the first 5 is the number of transfer functions, $D$ is the number of acoustic features in the dataset (22 for Oxford and 45 for Naranjo), and the final 5 is the number of potential classifiers. The first 5 dimensions are decoded as a one-hot scheme, that is, only one transfer function with the highest value in the corresponding dimension is selected. The first $D$ dimensions are binarized, via each optimizer's own transfer function, into a feature-selection mask over the $D$ acoustic features. The last 5 dimensions correspond to five candidate base classifiers, each with recent precedent as the wrapped classifier in its own evolutionary or metaheuristic feature-selection study: support vector machine (SVM, RBF kernel, calibrated via 5-fold Platt scaling for probability estimates [28]; paired with PSO-based feature selection for Alzheimer's disease diagnosis in [29]), random forest (RF, 100 trees; paired with an improved genetic algorithm for feature selection in [30]), extreme gradient boosting (XGBoost, 100 trees; paired with a multiobjective evolutionary algorithm for biomarker feature selection in [31]), $k$-nearest neighbors (KNN, $k = 5$; paired with physics-inspired metaheuristic feature selection in [32]), and logistic regression (LogReg; paired with particle swarm optimization for feature selection in chronic kidney disease diagnosis in [33]), chosen to span linear, instance-based, and both bagged and boosted tree-ensemble decision boundaries, so that the joint search can in principle favor whichever inductive bias best matches each dataset's feature space.

The last 5 dimensions are decoded the same way for every algorithm in this study: independent on/off switches, such that every classifier whose switch is on trains on the same selected feature subset, and the ensemble's out-of-fold prediction is the unweighted average of the active classifiers' predicted probabilities (i.e., equal-weight soft voting [34]), which lets a metaheuristic search directly over which base learners join a voting ensemble, rather than fixing the ensemble's membership in advance. Applying this multi-classifier ensemble decoding uniformly, rather than letting each baseline fall back to whatever single-classifier scheme its own binarization naturally produces, keeps the comparison in Section 5 isolated to each method's search strategy: every algorithm is optimizing over, and evaluated on, the identical solution space, so that a performance difference cannot be attributed to one method being handed a weaker or stronger classifier-selection scheme than another. Besides, if a particle ever decodes to an empty feature mask or an empty classifier mask, one bit is uniformly at random forced to be active to keep the candidate solution evaluable.

Figure 1 demonstrates this working pipeline end-to-end.

### *3.4. Transfer Functions for Binarization*

The swarm- and population-based metaheuristics used in this study were, in their original formulation, designed for continuous optimization: a particle's position is a real-valued vector, and the search dynamics, attraction toward better solutions, and so on, are defined in the continuous space. Joint feature-and-classifier selection is inherently discrete, since each dimension of a solution must resolve to a binary decision; that is, a given feature or classifier is either selected or not. A transfer function bridges this mismatch: it maps each continuous position value to a probability or a fixed threshold that determines the corresponding bit, so that an optimizer's continuous-space search dynamics can be reused unchanged while the solution actually evaluated is binary.

Five such transfer functions are adopted in this research, shown in Figure 2.

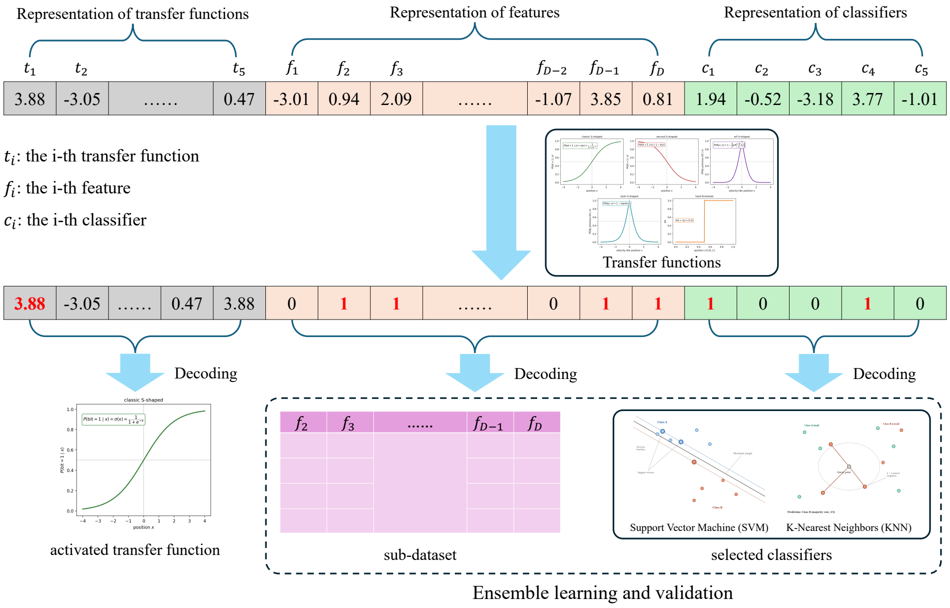


**Figure 1.** Worked example of the encoding and decoding pipeline. A continuous particle position of length $5 + D + 5$ splits into 5 transfer function scores ($t_1, \ldots, t_5$), $D$ feature scores ($f_1, \ldots, f_D$), and 5 classifier scores ($c_1, \ldots, c_5$). Each segment passes through a transfer function to produce a binary string; thresholded-off features/classifiers (0) are excluded, and the remainder (1, highlighted) are decoded into a feature mask, here selecting the sub-dataset restricted to columns $f_2, f_3, \ldots, f_{D-1}, f_D$, and a classifier mask, here selecting SVM and KNN. Every classifier in the mask trains on the same feature subset and is combined by equal-weight soft voting before 5-fold subject-grouped cross-validation.

The classic stochastic S-shaped transfer function adopts the logistic sigmoid to give the probability that a bit is set, as formulated in Eq. (1).

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \qquad \Pr[\text{bit} = 1] = \sigma(x), \tag{1}$$

so that selection probability increases monotonically with $x$; this remains an active choice of binarization scheme in recent metaheuristic feature-selection studies [13].

A second S-shaped variant, following Hashemi et al. [17], reuses the same logistic sigmoid but decodes in the opposite direction from Eq. (1), which is defined in Eq. (2).

$$\Pr[\text{bit} = 1] = 1 - \sigma(x), \tag{2}$$

Another V-shaped curve is also defined in [17] that instead governs whether the previous bit flips, with flip probability in Eq. (3).

$$\Pr[\text{bit flips}] = 1 - V(x), \qquad V(x) = \left| \operatorname{erf}\!\left( \frac{\sqrt{\pi}}{2} x \right) \right|, \tag{3}$$

which peaks at $x = 0$ (where $V(0) = 0$) and decays toward 0 as $|x|$ grows (where $V(x) \to 1$).

Besides, a tanh V-shaped curve, following Mirjalili and Lewis's [35] own V-shaped family, uses the hyperbolic tangent in place of the error function:

$$\Pr[\text{bit flips}] = 1 - V_2(x), \qquad V_2(x) = |\tanh(x)|, \tag{4}$$

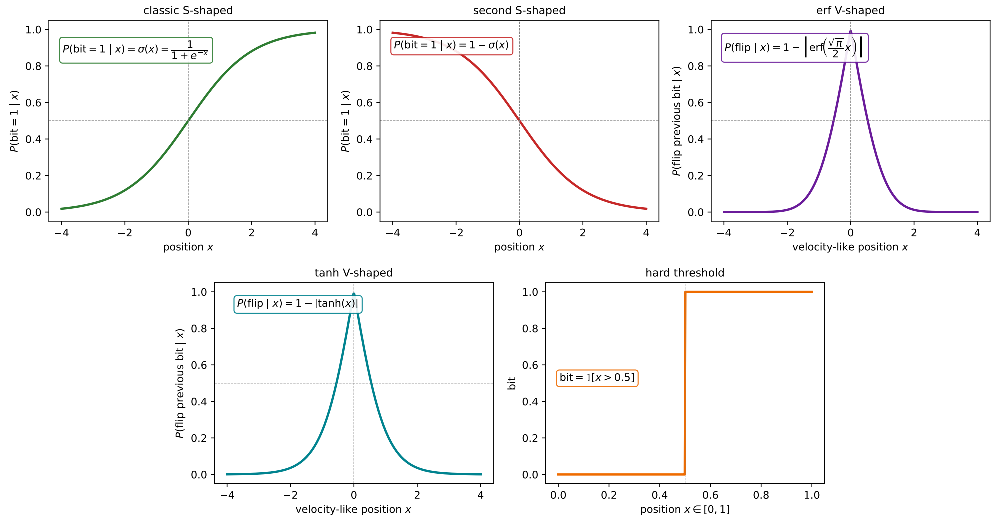


**Figure 2.** The five transfer functions used in this study: Top row, left to right: classic S-shaped (Eq. (1)) [13]; second S-shaped (Eq. (2)) [17]; erf V-shaped (Eq. (3)) [17]. Bottom row, centered under the top row: tanh V-shaped (Eq. (4) [35]; hard threshold (Eq. (5)) [20], used natively in $[0, 1]$.

with the same flip-the-previous-bit mechanic and the same peaks-at-zero, decays-as-$|x|$-grows shape as Eq. (3), merely a cheaper closed form. Finally, a hard-threshold binarization [20] applies directly to positions that live natively in $[0, 1]$, with no stochastic component:

$$\text{bit} = \begin{cases} 1 & x > 0.5, \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

### *3.5. Fitness Function*

Every candidate solution is scored by a single scalar fitness, combining diagnostic performance with a feature-parsimony penalty in Eq. (6):

$$\text{fitness} = \omega \cdot (1 - \text{BalAcc}) + (1 - \omega) \cdot \frac{|S|}{D} \tag{6}$$

where BalAcc is the out-of-fold balanced accuracy (the average of sensitivity and specificity, robust to Oxford's class imbalance [36]) achieved by the decoded feature subset and ensemble of classifiers, $|S|$ is the number of selected features, $D$ is the total number of available features, and $\omega = 0.9$ weights diagnostic performance nine times as heavily as feature count, as suggested in [37]. This value ensures a modest reduction in feature count is rewarded only when it does not come at a material cost to balanced accuracy. Additionally, lower fitness is better, and every optimizer in this study is implemented as a minimizer.

### *3.6. Our Proposed Pipeline: The Joint Optimization via Competitive Swarm Optimizer*

The contribution of this study is not a new metaheuristic operator but a new paradigm for handling metaheuristic-based feature selection efficiently. As reviewed in Section 2.3, published metaheuristic pipelines for PD voice screening almost universally treat the transfer function, the feature subset, and the classifier configuration as three separate decisions, of which only the feature subset is determined via metaheuristic search, while the other two components are configured by hand according to expert judgment. This section describes the pipeline that instead searches all three jointly, within the single $5 + D + 5$-dimensional encoding of Section 3.3 and the fitness of Section 3.5, using the Competitive Swarm Optimizer (CSO) [38] as the search engine that explores this space generation by generation.

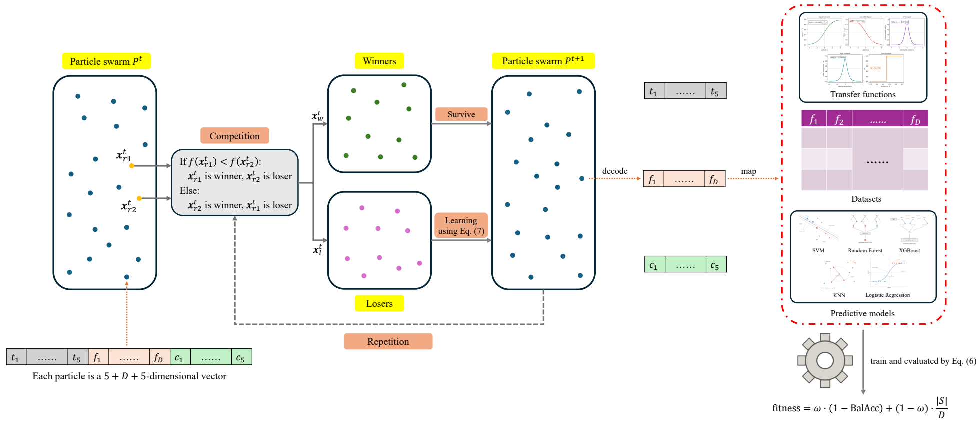


**Figure 3.** One full generation of the proposed pipeline.

#### 3.6.1. Basic CSO

The Competitive Swarm Optimizer (CSO) was originally proposed by Cheng and Jin in 2015 [38], which departs from the attractor-based update rule common to PSO, where every particle is pulled toward the personal best and global best solution. CSO instead structures each generation as a set of pairwise competitions: only the loser of each pair updates and learns from the winner particle and the centroid position. This design is intended to preserve swarm diversity and allocate the computational budgets to the promising solutions. Concretely, in each generation, the even-numbered population of $N$ particles is randomly paired into $N/2$ one-on-one competitions. Within each pair, the fitter particle, denoted as $\boldsymbol{v}_w^t$, passes to the next generation unchanged, while the weaker particle, denoted as $\boldsymbol{v}_l^t$, updates its velocity and position toward the winner and the centroid of the swarm, as described in Eq. (7):

$$
\begin{aligned}
\boldsymbol{v}_l^{t+1} &= \boldsymbol{R}_1 \cdot \boldsymbol{v}_l^t + \boldsymbol{R}_2 \cdot (\boldsymbol{x}_w^t - \boldsymbol{x}_l^t) + \varphi \cdot \boldsymbol{R}_3 \cdot (\bar{\boldsymbol{x}}^t - \boldsymbol{x}_l^t) \\
\boldsymbol{x}_l^{t+1} &= \boldsymbol{x}_l^t + \boldsymbol{v}_l^{t+1}
\end{aligned} \tag{7}
$$

where $\boldsymbol{x}_w^t$ and $\boldsymbol{x}_l^t$ are the winner's and loser's positions, respectively, $\bar{x}$ is the centroid of the swarm, $\boldsymbol{R}_1$, $\boldsymbol{R}_2$, and $\boldsymbol{R}_3$ are independent uniform random vectors, and $\phi$ is CSO's social-term control coefficient, where the sensitivity analysis for $\phi$ in this specific joint optimization task is provided in Section 4.3.

#### 3.6.2. The Implementation of Our Proposed Framework

Figure 3 traces one full generation of the proposed pipeline end-to-end, tying together the encoding of Section 3.3, the transfer-function candidates of Section 3.4, the CSO update of Section 3.6.1, and the fitness of Section 3.5 into the single procedure that this study actually runs.

Each particle in the swarm $P^t$ is a $5 + D + 5$-dimensional vector $[t_1, \ldots, t_5, f_1, \ldots, f_D, c_1, \ldots, c_5]$. Every generation, particles are randomly paired and compared based on the competition mechanism in CSO; each pair's winner survives unchanged into $P^{t+1}$, while its loser's entire position is updated by Eq. (7) before being decoded and re-evaluated. Decoding a position for the updated particle individual consists of three steps:

1. **Transfer-function selection**: The leading segment $[t_1, \ldots, t_5]$ is argmaxed to pick one of the 5 candidate transfer functions without a binarization step.
2. **Feature and classifier mapping**: The selected candidate binarizes the trailing $[f_1, \ldots, f_D, c_1, \ldots, c_5]$ segment based on the specific transfer function. The resulting $D$-bit feature mask selects which of the dataset's $D$ acoustic-feature columns are retained (map in Figure 3); the resulting 5-bit classifier mask selects which of the five candidate classifiers are included in that particle's ensemble.

**Table 1.** Hyperparameters of algorithms.

| Algorithm | Hyperparameters |
|---|---|
| BGWO | $a : 2 \to 0$ |
| HybridGWO | WOA: $b = 1.0$; GWO: $a : 2 \to 0$ |
| QMFO | $a_1 = 1.0$, $a_2 = 1.5$, $\beta = 2.0$, dance $= 5.0$, |
| | $fl = 1.0$, rotation step $= 0.15$ |
| BPSO | $w = 0.7$, $c_1 = 2.5$, $c_2 = 1.6$ |
| BBOA | $c = 0.01$, $a \in [0.1, 0.3]$, $p = 0.6$ |
| CSO | $\varphi = 0$ (As recommended in [38] that when $N < 100$, $\varphi = 0$) |

3. **Training and evaluation**: The selected classifiers are trained on the selected feature subset under the subject-grouped cross-validation protocol of Section 3.2 and combined into an equal-weight soft-voting ensemble; the resulting out-of-fold balanced accuracy and feature ratio are combined into the scalar fitness computed by Eq. (6).

The resulting fitness value is what determines that particle's winner/loser status the next time it is paired, closing the loop shown at the bottom of Figure 3: decode, map, train, and evaluate happen every time a loser updates, generation after generation, for every particle in the swarm.

### *3.7. Comparison Methods*

Five metaheuristic feature-selection methods from recently published papers, specifically focus on PD diagnosis, are re-implemented to enable a controlled, like-for-like comparison against the proposed framework: binary grey wolf optimizer (BGWO) and a whale-optimization-then-grey-wolf refinement cascade (HybridGWO) [24], mayfly algorithm with a quantum-rotation-gate mutation operator (QMFO)[21], and binary particle swarm optimization (BPSO) as well as butterfly optimization algorithm (BBOA) with a hybrid S-shaped or V-shaped transfer function [17].

Every optimizer is compared under an equal fitness-evaluation budget ($FE_{max}$) rather than an equal generation count, since different algorithms consume different numbers of fitness evaluations per generation (e.g., the hybrid S-shaped/V-shaped transfer function used by BPSO/BBOA evaluates two binarization candidates per individual per generation; mHGS's production and escaping operators each conditionally trigger additional evaluations). A population size of 30 and a fitness-evaluation budget of $FE_{max} = 3000$ per run are used throughout. Each algorithm-dataset combination is run with multiple independent seeds (seed $= 2026 + 50 \times i$, where $i$ is the run index in $\{1, 2, \ldots, 50\}$, so seeds do not overlap across runs). This fair and detailed comparison against the seven baselines, carried out across multiple environments, uses enough independent runs to support the statistical testing described in Section 4.

### *3.8. Evaluation Metrics*

For every run, the out-of-fold predictions used to compute fitness are also used to report a full panel of standard diagnostic-performance metrics, computed once per run from the same confusion matrix, whose entries are true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), with the Parkinsonian class taken as positive:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \tag{9}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{10}$$

$$\text{BalAcc} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \tag{11}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{12}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{13}$$

$$\text{ROC-AUC} = \Pr\left(\hat{p}(x^+) > \hat{p}(x^-)\right) \tag{14}$$

where $\hat{p}(x)$ is the predicted probability of the Parkinsonian class for a recording $x$, and $x^+$, $x^-$ are independently drawn Parkinsonian and healthy-control recordings, respectively. That is, ROC-AUC, formulated in Eq. (14), is the probability that the model ranks a randomly drawn positive case above a randomly drawn negative one, equivalently the area under the curve traced by the true positive rate versus the false positive rate as the decision threshold varies, unlike Eqs. (8) to (13), which are all evaluated at the single decision threshold (0.5) used elsewhere in this study. Sensitivity and specificity are reported alongside the more ML-conventional accuracy/F1/AUC metrics because they are the standard vocabulary for evaluating a clinical screening tool: sensitivity in Eq. (9) quantifies the fraction of true PD cases the pipeline would flag (missed cases have direct clinical cost), while specificity in Eq. (10) quantifies the fraction of healthy individuals correctly spared unnecessary follow-up. BalAcc in Eq. (11) is the same quantity used inside the fitness function, so the primary search objective and the headline reported metric are one and the same, not two different notions of accuracy. Feature-selection ratio ($|S|/D$) and the identity of the selected feature subset and active classifier(s) are also recorded for every run, supporting the feature-interpretability analysis in Section 5.

## 4. Results

This section compares the proposed CSO pipeline against the five baselines from Section 3.7 on both datasets, under the equal fitness-evaluation budget and multi-seed protocol described.

### 4.1. Comparison on the Oxford Dataset

Table 2 summarizes the mean ± standard deviation of every reported metric on the Oxford dataset across all independent runs of each algorithm.

Figure 4 shows the mean best-fitness trajectory, as computed in Eq. (6), against the number of fitness evaluations consumed, averaged over all independent runs of each algorithm.

Figure 5 shows the distribution of out-of-fold balanced accuracy, as computed in Eq. (11), achieved by each algorithm across its independent runs.

Figure 6 reports the mean and standard deviation of balanced accuracy, F1, as computed in Eq. (13), and ROC-AUC, as computed in Eq. (14), for each algorithm.

Figure 7 reports the mean and standard deviation of the selected feature ratio, $|S|/D$ in Eq. (6), for each algorithm.

**Table 2.** Mean ± standard deviation of balanced accuracy, F1, ROC-AUC, and selected feature ratio on the Oxford dataset ($n = 20$ runs per algorithm). Bold marks the best value per column.

| Algorithm | Balanced Accuracy | F1 | ROC-AUC | Feature Ratio |
|---|---|---|---|---|
| BGWO | $0.801 \pm 0.020^{\dagger}$ | $0.913 \pm 0.009$ | $0.846 \pm 0.040$ | $0.227 \pm 0.064$ |
| HybridGWO | $\mathbf{0.815 \pm 0.021}$ | $\mathbf{0.920 \pm 0.010}$ | $\mathbf{0.863 \pm 0.032}$ | $0.255 \pm 0.104$ |
| QMFO | $0.787 \pm 0.032^{\ddagger}$ | $0.909 \pm 0.015$ | $0.855 \pm 0.035$ | $0.300 \pm 0.076$ |
| BPSO | $0.798 \pm 0.029^{\dagger}$ | $0.912 \pm 0.012$ | $0.860 \pm 0.038$ | $0.248 \pm 0.141$ |
| BBOA | $0.796 \pm 0.023^{\dagger}$ | $0.910 \pm 0.015$ | $0.852 \pm 0.040$ | $0.230 \pm 0.137$ |
| CSO | $0.810 \pm 0.025$ | $0.918 \pm 0.013$ | $0.861 \pm 0.037$ | $\mathbf{0.225 \pm 0.130}$ |

$^{\dagger}p < 0.05$ vs. CSO; $^{\ddagger}p < 0.05$ vs. CSO and survives a Bonferroni-corrected threshold of $\alpha = 0.01$ for the five simultaneous comparisons (paired Wilcoxon signed-rank test on balanced accuracy, matched by seed; Section 5.2). Unmarked rows (HybridGWO) are not significantly different from CSO.

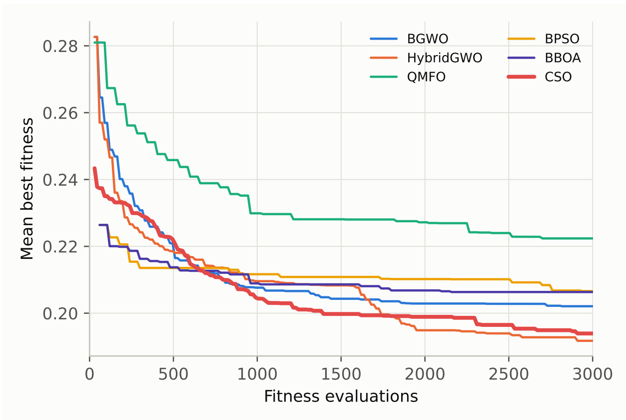


**Figure 4.** Mean best-fitness convergence on the Oxford dataset. Lower is better; CSO is highlighted.

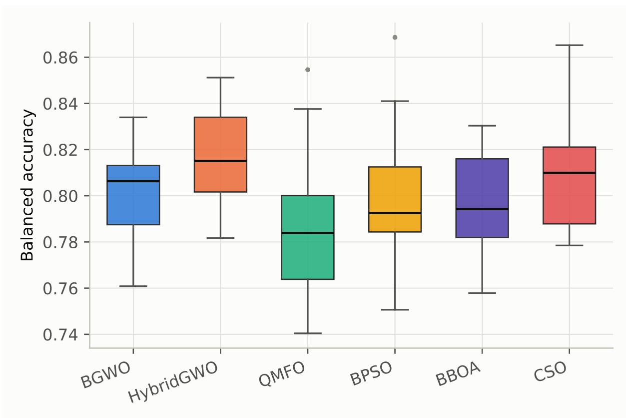


**Figure 5.** Balanced-accuracy distribution across runs on the Oxford dataset.

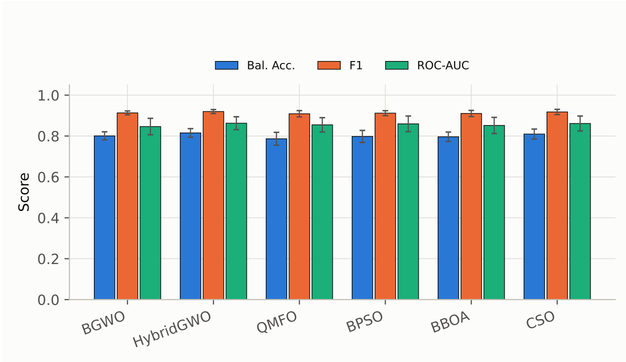


**Figure 6.** Mean ± standard deviation of balanced accuracy, F1, and ROC-AUC on the Oxford dataset.

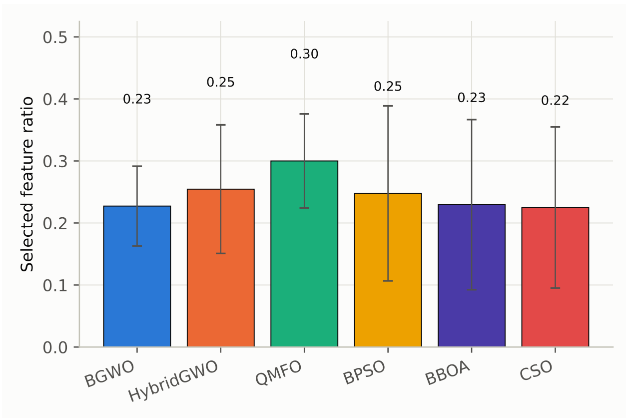


**Figure 7.** Mean ± standard deviation of the selected feature ratio on the Oxford dataset.

### *4.2. Comparison on the Naranjo Dataset*

Table 3 summarizes the mean $\pm$ standard deviation of every reported metric on the Naranjo dataset across all independent runs of each algorithm.

Figure 8 shows the mean best-fitness trajectory, as computed in Eq. (6), against the number of fitness evaluations consumed, averaged over all independent runs of each algorithm.

Figure 9 shows the distribution of out-of-fold balanced accuracy, as computed in Eq. (11), achieved by each algorithm across its independent runs.

Figure 10 reports the mean and standard deviation of balanced accuracy, F1, as computed in Eq. (13), and ROC-AUC, as computed in Eq. (14), for each algorithm.

Figure 11 reports the mean and standard deviation of the selected feature ratio, $|S|/D$ in Eq. (6), for each algorithm.

**Table 3.** Mean ± standard deviation of balanced accuracy, F1, ROC-AUC, and selected feature ratio on the Naranjo dataset ($n = 20$ runs per algorithm). Bold marks the best value per column.

| Algorithm | Balanced Accuracy | F1 | ROC-AUC | Feature Ratio |
|---|---|---|---|---|
| BGWO | $0.840 \pm 0.009^{\dagger}$ | $0.836 \pm 0.009$ | $\mathbf{0.864 \pm 0.015}$ | $0.288 \pm 0.052$ |
| HybridGWO | $0.843 \pm 0.007$ | $0.840 \pm 0.008$ | $0.861 \pm 0.014$ | $0.310 \pm 0.059$ |
| QMFO | $0.833 \pm 0.008^{\ddagger}$ | $0.831 \pm 0.008$ | $0.858 \pm 0.013$ | $0.377 \pm 0.062$ |
| BPSO | $0.830 \pm 0.009^{\ddagger}$ | $0.828 \pm 0.009$ | $0.856 \pm 0.015$ | $0.234 \pm 0.097$ |
| BBOA | $0.827 \pm 0.008^{\ddagger}$ | $0.824 \pm 0.007$ | $0.859 \pm 0.013$ | $\mathbf{0.177 \pm 0.051}$ |
| CSO | $\mathbf{0.847 \pm 0.012}$ | $\mathbf{0.843 \pm 0.012}$ | $0.862 \pm 0.020$ | $0.261 \pm 0.066$ |

$^{\dagger}p < 0.05$ vs. CSO; $^{\ddagger}p < 0.05$ vs. CSO and survives a Bonferroni-corrected threshold of $\alpha = 0.01$ for the five simultaneous comparisons (paired Wilcoxon signed-rank test on balanced accuracy, matched by seed; Section 5.2). Unmarked rows (HybridGWO) are not significantly different from CSO.

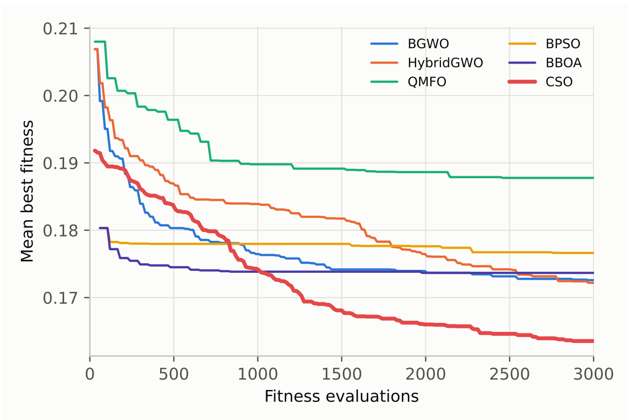


**Figure 8.** Mean best-fitness convergence on the Naranjo dataset. Lower is better; CSO is highlighted.

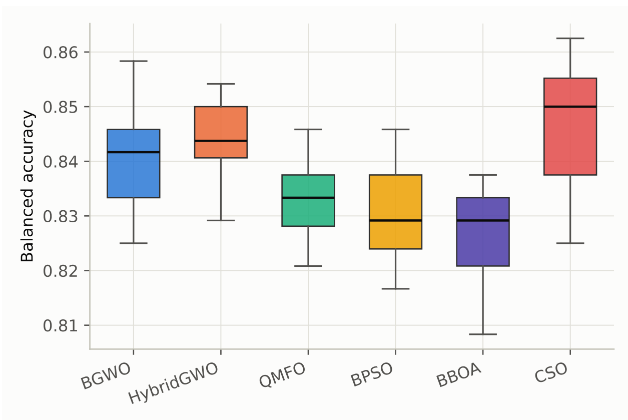


**Figure 9.** Balanced-accuracy distribution across runs on the Naranjo dataset.

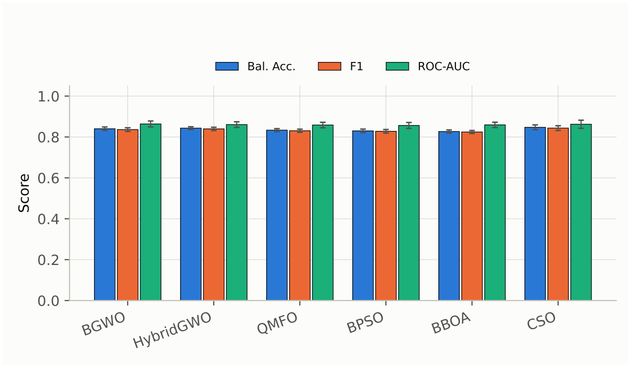


**Figure 10.** Mean ± standard deviation of balanced accuracy, F1, and ROC-AUC on the Naranjo dataset.

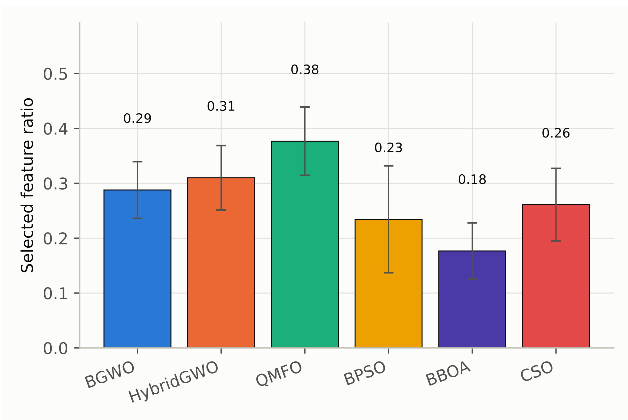


**Figure 11.** Mean ± standard deviation of the selected feature ratio on the Naranjo dataset.

### *4.3. Sensitivity Analysis of the CSO Social-Term Coefficient $\varphi$*

As noted in Table 1, the proposed pipeline uses $\varphi = 0$ throughout the comparison of Sections 4.1 and 4.2, since Cheng and Jin's own formula for $\varphi$ (Eq. (7)) is exactly 0 at this study's population size ($m = 30 \leq 100$). To check whether this formula-mandated value is actually a good one for this joint optimization task, rather than simply the value the original CSO paper happens to prescribe at this scale, $\varphi$ is swept over $\{0.00, 0.05, 0.10, \ldots, 0.50\}$ with every other setting unchanged (population size 30, $FE_{max} = 3000$, 20 independent seeds per $\varphi$ value per dataset), overriding the formula-derived value with each fixed candidate in turn.

Figures 12 and 13 show mean $\pm$ standard deviation of the best fitness (Eq. (6)) attained under each $\varphi$, on Oxford and Naranjo respectively; the circled marker highlights the best-performing value. On both datasets, $\varphi = 0$ attains the lowest mean best fitness of the eleven values swept (Oxford: 0.194, versus 0.201–0.214 for $\varphi > 0$; Naranjo: 0.163, versus 0.167–0.175 for $\varphi > 0$), with fitness drifting upward, i.e., worse, as $\varphi$ increases from 0.

Figures 14 and 15 decompose this into the balanced-accuracy term of the fitness function. $\varphi = 0$ attains the highest mean balanced accuracy on both datasets (Oxford: 0.810; Naranjo: 0.847), and on Naranjo the decline as $\varphi$ increases is close to monotonic.

Figures 16 and 17 report the selected feature ratio, $|S|/D$, against $\varphi$. Unlike best fitness and balanced accuracy, this term shows no clear trend with $\varphi$ on either dataset (Oxford's minimum falls at $\varphi = 0.05$, Naranjo's at $\varphi = 0$), consistent with the fitness function weighting feature ratio an order of magnitude less than accuracy ($\omega = 0.9$ in Eq. (6)), so its sample noise across only 20 seeds is not expected to resolve a clean trend the way the accuracy-dominated fitness does.

Overall, this sensitivity analysis confirms that $\varphi = 0$ is not merely a value CSO's own formula happens to prescribe at this study's population size, but the empirically best-performing choice on both datasets: reintroducing a nonzero mean-position pull into the loser update never improves, and on Naranjo steadily worsens, the joint search's fitness and accuracy.

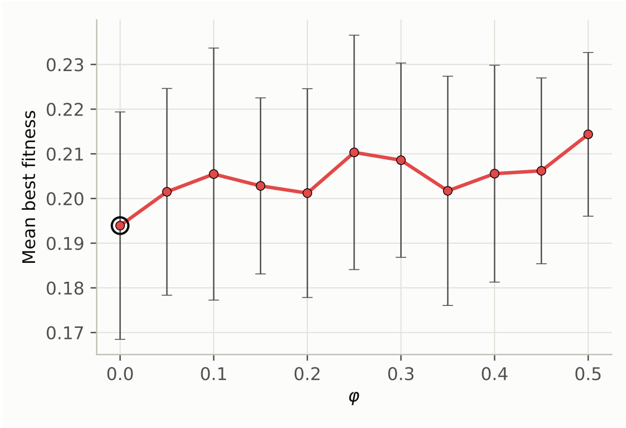


**Figure 12.** Mean ± standard deviation of best fitness against $\varphi$ on the Oxford dataset. The circled marker is the best-performing $\varphi$.

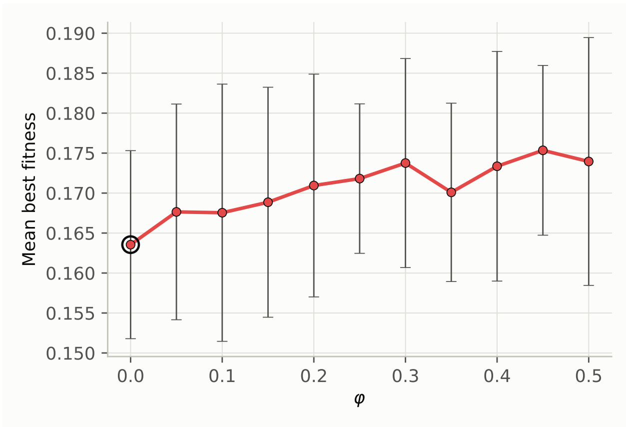


**Figure 13.** Mean ± standard deviation of best fitness against $\varphi$ on the Naranjo dataset. The circled marker is the best-performing $\varphi$.

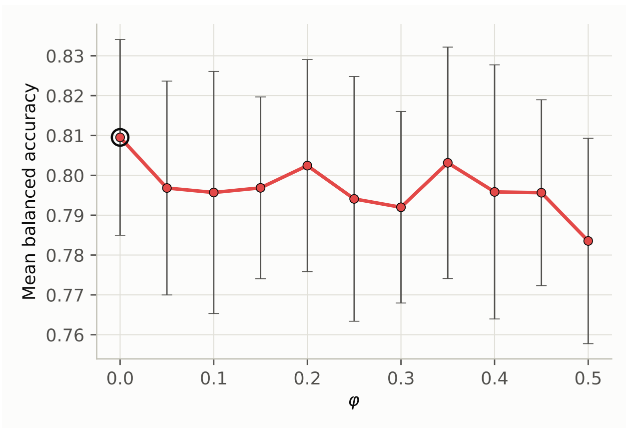


**Figure 14.** Mean ± standard deviation of balanced accuracy against $\varphi$ on the Oxford dataset. The circled marker is the best-performing $\varphi$.

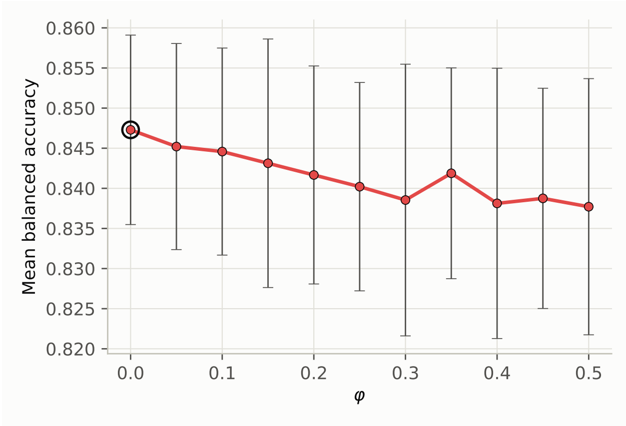


**Figure 15.** Mean ± standard deviation of balanced accuracy against $\varphi$ on the Naranjo dataset. The circled marker is the best-performing $\varphi$.

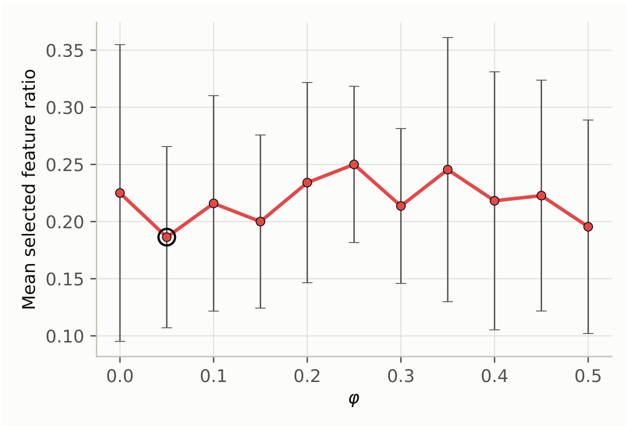


**Figure 16.** Mean ± standard deviation of the selected feature ratio against $\varphi$ on the Oxford dataset. The circled marker is the best-performing $\varphi$.

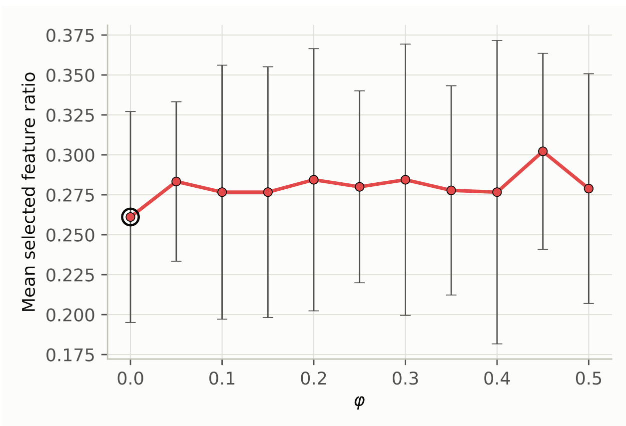


**Figure 17.** Mean ± standard deviation of the selected feature ratio against $\varphi$ on the Naranjo dataset. The circled marker is the best-performing $\varphi$.

## 5. Discussion

### 5.1. On BPSO and BBOA's Early-Generation Fitness Advantage

Figures 4 and 8 show BPSO and BBOA reaching a markedly better mean best-fitness than every other method within the very first recorded evaluations, before plateauing early and eventually being overtaken by BGWO, HybridGWO, and the proposed CSO pipeline. This is not the result of an informed or non-random initialization scheme: every algorithm in this study, including BPSO and BBOA, draws its initial population from the same uniform distribution with no warm-start, opposition-based, or chaotic-map seeding. The advantage instead comes from the hybrid S-shaped/V-shaped transfer function that Hashemi et al. [17] specify for BPSO and BBOA (Section 3.7): for every individual, at every generation including the very first, the same continuous position is binarized twice, once under the S-shaped rule of Eq. (2) and once under the V-shaped rule of Eq. (3), both candidates are evaluated, and whichever achieves the lower fitness is kept. Because a best-of-two selection cannot score worse, in expectation, than a single draw, BPSO and BBOA's reported fitness at any generation is a best-of-two order statistic rather than a single sample from the same distribution the other baselines and the proposed method draw from, which explains why their initial population already looks stronger despite an identical, uninformed random start.

This mechanism also means BPSO and BBOA spend two fitness evaluations per individual per generation wherever the other five methods spend one, so under the shared evaluation budget $FE_{max}$ (Section 3.7) they complete roughly half as many generations. Their early lead is therefore a one-time gain from the transfer function's built-in selection pressure rather than evidence of a more effective search or exploitation mechanism; it does not compound across generations the way the winner/loser competitive update of CSO (Eq. (7)) or the GWO family's convergence-controlled encircling does, which is consistent with BPSO and BBOA's curves flattening earliest in Figures 4 and 8 while every other method continues to improve.

### 5.2. On the Comparison Between CSO and the Baselines

CSO's advantage over the five baselines is dataset-dependent rather than uniform. On Naranjo, it attains the highest mean balanced accuracy of all six methods (0.847, versus 0.827–0.843 for the baselines; Figure 9), whereas on Oxford it ranks second (0.810), behind HybridGWO's 0.815 (Figure 5). A paired Wilcoxon signed-rank test on balanced accuracy, matched by seed across algorithms ($n = 20$ per comparison), confirms this asymmetry. On Oxford, CSO significantly outperforms BGWO ($p = 0.048$), QMFO ($p = 0.007$), and BPSO ($p = 0.036$), and is on the margin against BBOA ($p = 0.050$); only its comparison with QMFO survives a Bonferroni-corrected threshold of $\alpha = 0.01$ for the five simultaneous comparisons. Against HybridGWO, CSO is statistically indistinguishable ($p = 0.51$, a near-even 10/20 pairwise split) – the only baseline not clearly outperformed on either dataset. On Naranjo, CSO significantly outperforms all five baselines (BGWO $p = 0.028$, HybridGWO $p = 0.079$, QMFO $p < 0.001$, BPSO $p < 0.001$, BBOA $p < 0.001$), of which the comparisons against QMFO, BPSO, and BBOA remain significant under the same Bonferroni-corrected threshold.

This pattern is consistent with the higher dimensionality of Naranjo's encoding: at $D = 45$ acoustic features, versus $D = 22$ for Oxford, the joint search space of Section 3.3 is roughly twice as large, giving CSO's per-particle searched transfer function – letting each candidate solution pick whichever of five binarization rules (Section 3.6.2) suits its own region of the space, rather than every particle sharing one fixed rule – correspondingly more room to specialize. On the smaller Oxford space, this adaptive advantage is thinner, and HybridGWO's fixed WOA-to-GWO cascade schedule (Table 1) is competitive without it.

CSO's efficiency profile (Figures 7 and 11) also differs by dataset. On Oxford, it selects the smallest mean feature ratio of all six methods (0.225, versus 0.227–0.300 for the baselines) while simultaneously improving accuracy, that is, no accuracy-parsimony trade-off is paid.

On Naranjo, it selects more features on average (0.261) than BBOA (0.177) or BPSO (0.234), trading some parsimony for its accuracy lead. Because the fitness function (Eq. (6)) weights balanced accuracy nine times as heavily as feature ratio ($\omega = 0.9$), this is the expected behavior rather than an inconsistency: whenever a larger feature subset yields a material accuracy gain, the search is designed to keep it. QMFO selects the largest feature subsets on both datasets (0.300 on Oxford, 0.377 on Naranjo) without a corresponding accuracy advantage, suggesting its search dynamics are comparatively less effective than the other methods at pruning uninformative features under this joint encoding.

## 6. Conclusion

**Author Contributions:**

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**Institutional Review Board Statement:** Not applicable

**Informed Consent Statement:** Not applicable

**Data Availability Statement:** The datasets analyzed in this study are publicly available in the UCI Machine Learning Repository: the Oxford Parkinson's Disease Voice Dataset at https://archive.ics.uci.edu/dataset/174/parkinsons, and the Naranjo Replicated Acoustic Features Dataset at https://archive.ics.uci.edu/dataset/489/parkinson+dataset+with+replicated+acoustic+features. No new data were collected for this study.

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